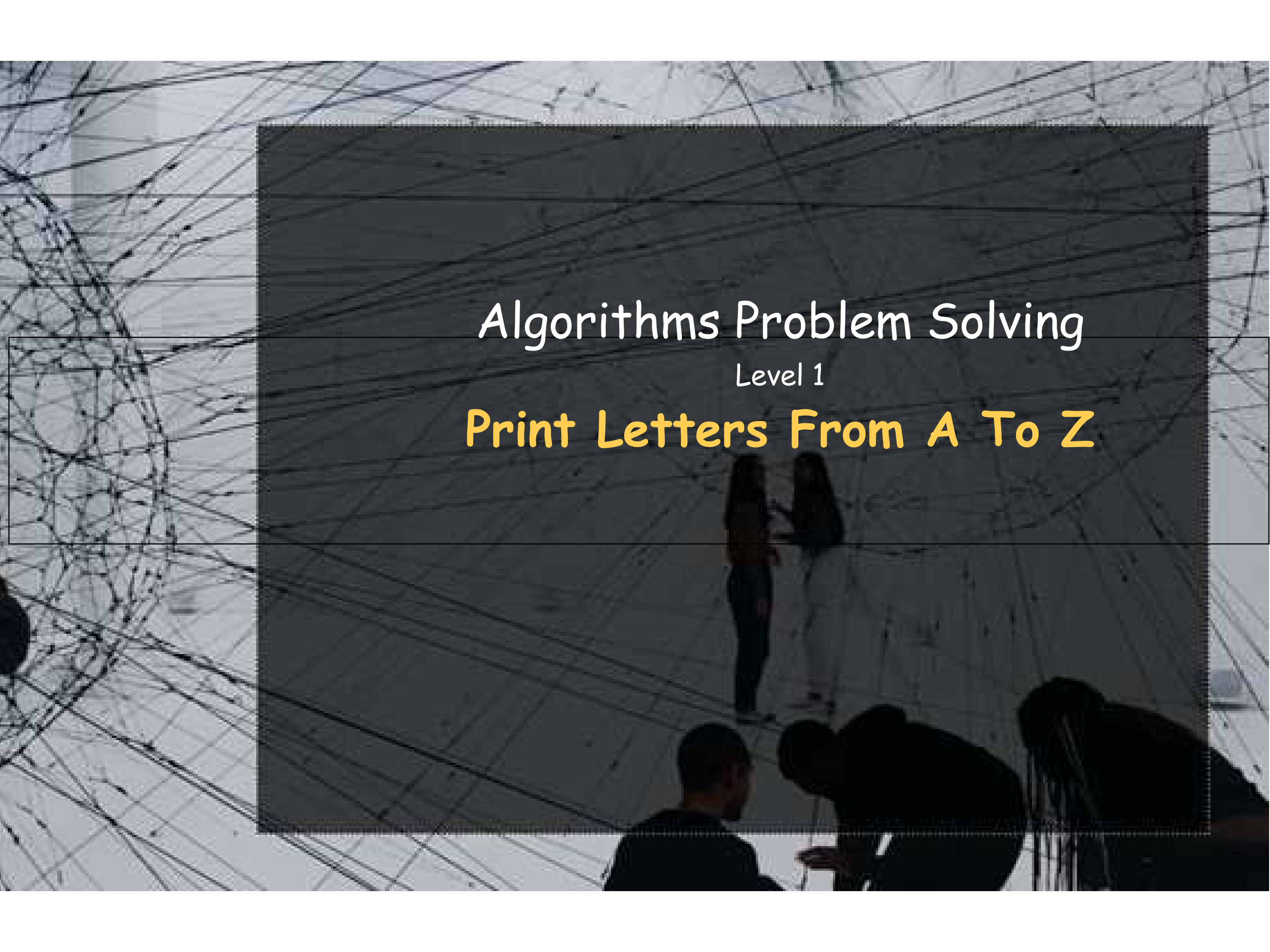


Algorithms Problem Solving

Level 1

Print Letters_From A To
Z

The background of the slide is a photograph of a modern building's interior. It features a large, curved glass ceiling that allows natural light to filter through, creating a pattern of light and shadow on the floor. In the lower part of the image, the silhouettes of several people are visible, some standing and some sitting, suggesting a collaborative workspace or a public area. The overall color palette is muted, with greys, blues, and the warm tones of the light coming through the glass.

Algorithms Problem Solving

Level 1

Print Letters From A To Z

Definition

- ◆ This program prints all uppercase letters of the English alphabet from `A` to `Z`.
- ◆ The program uses a `char` variable and a `while` loop to move through the letters.
- ◆ The `++` operator increases the character to the next letter.

Explanation

- ◆ The variable ``letter`` starts with the character ``'A'``.
- ◆ The condition ``letter <= 'Z'`` keeps the loop running while the current character is within the alphabet range.
- ◆ ``cout << letter << " ";`` prints the current letter.
- ◆ ``letter++`` moves to the next character.
- ◆ When ``letter`` becomes greater than ``'Z'``, the loop stops.
- ◆ ``printAlphabet()`` contains the complete printing logic, while ``main()`` only calls the function.

Steps in main()

1. Set `letter` to `'A'`.
2. Check whether `letter <= 'Z'`.
3. Print the current letter.
4. Increment `letter`.
5. Repeat until the letter passes `'Z'`.
6. Print a new line.

💡 Example

◆ The loop starts with:
`A`

◆ Then it continues:
`B C D E ...`

◆ The final letter is:
`Z`

➡ Result

A B C D E F G H I J K L M N O P Q R S T U V W X Y Z

Note

- ◆ Characters have numeric character codes, so incrementing a `char` moves to the next character in the character encoding.
- ◆ For the English uppercase alphabet, `'A'` through `'Z'` are stored consecutively.

⊘ Warning

◆ This approach relies on the alphabet characters being consecutive in the character encoding, which is guaranteed for the basic execution character set used by standard C++ implementations.



Comparison

`while` Loop

- ✓ Starts at ``'A'``.
- ✓ Continues while the condition is true.
- ✓ Suitable when the loop progresses character by character.

Manual Printing

- ✓ Each letter would have to be written separately.
- ✓ Produces much more repetitive code.
- ✓ Does not demonstrate the benefit of a loop.

✦✦ Summary

- ✓ A `char` variable can be used to iterate through letters.
- ✓ `'A'` to `'Z'` represents the uppercase English alphabet.
- ✓ `while` repeats the printing operation until the condition becomes false.
- ✓ `letter++` moves to the next character.

Thank you 😊